



I N V E N T W O O D

SUPERWOOD for Data Centers

Companion to the SUPERMILLS Investor Overview
September 2026

THE THESIS

We help data centers decarbonize and improve their impact on communities, while they accelerate SUPERWOOD's journey from premium skins into *structural* applications

01

The buyer is short of steel and already building with wood

Data centers are among the fastest-growing buyers of structural steel. Customers report backlogs and shortages, and Microsoft and Meta already build data-center structures in mass timber.

02

We turbocharge the wood they already use

Mass timber replaces concrete floors. SUPERWOOD, stronger than A36 steel in tension at one-sixth the weight, adds the steel: members, skins and screens — and strengthens the mass timber itself. It ships today from SuperMill One; truss design and mass-timber enhancement are under way.

03

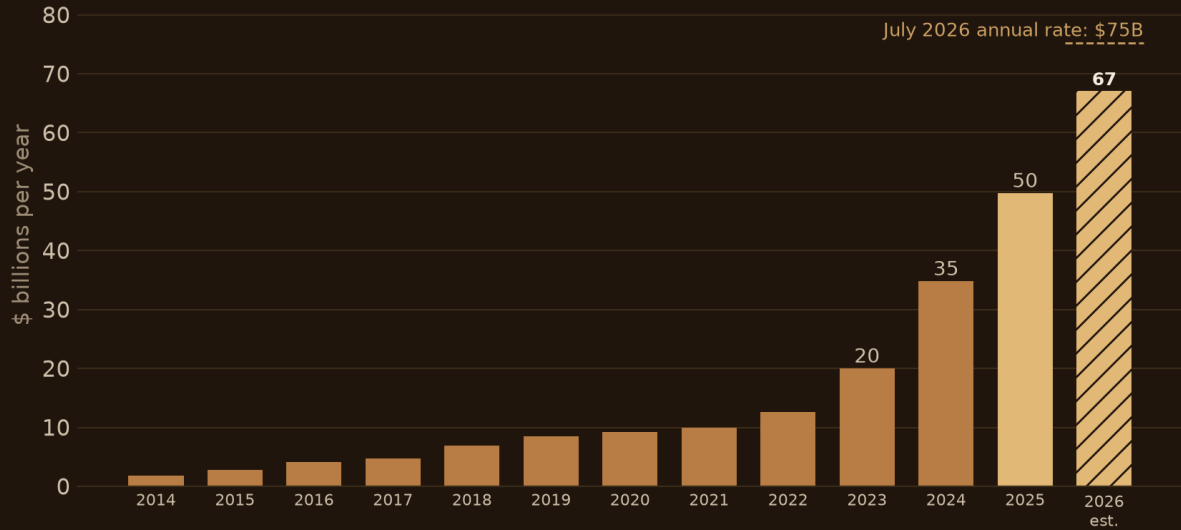
One basis-of-design win is SuperMill Two-scale demand

A gigawatt data center's skins alone are one to three years of SuperMill One's output; its structure is a year or more of SuperMill Two. Long term: prefabricated envelopes and, eventually, foundations.

Note 1

Industry Growth

US data center construction is running at *\$75 billion a year*, up 57% in twelve months.



HOW IT IS PROJECTED TO GROW

● 66 GW under construction in North America

JLL, midyear 2026 — more electricity than Germany uses. At the published 50–100 tons of structural steel per MW, that pipeline alone is 3.3–6.6 million tons of structural steel.

● Global capacity roughly triples by 2030

McKinsey: 82 GW in 2025 to about 219 GW in 2030, AI workloads driving 71% of the growth. JLL: 103 GW to 200 GW.

● Global data center capex passes \$1 trillion in 2026

Dell'Oro: up 57% in 2025, above \$1 trillion in 2026, \$1.7 trillion by 2030 — mostly servers and chips; the buildings are the Census share on the left.

Industry Challenged by Communities

Community pushback now blocks or delays more data center projects than ever.



COMPLAINT

Aesthetics

Windowless boxes on farmland. Counties answer with setbacks and design standards.

SUPERWOOD Facades, fences and screens that read as architecture.



COMPLAINT

Noise

Fans, chillers, generator tests. Property-line decibel limits are spreading.

SUPERWOOD Acoustic screens and barriers; quieter fan enclosures and ductwork.

Concept renderings

75 projects, about \$130 billion, blocked or delayed in one quarter

Q1 2026, as much as all of 2025. 833 opposition groups in 49 states.

Rules are following

Henry County, Virginia: 1,000 ft setbacks, 50 dBA at the property line.

We help with these two. *We can go much further*: into the structure of the building and what is inside it.

Note 3

Hyperscalers are already building structures with wood. We *turbocharge* wood.



Two Northern Virginia datacenters with cross-laminated timber floors on a steel frame — about 35% less embodied carbon than conventional steel construction, 65% less than precast. Gensler; Thornton Tomasetti.

Microsoft Source, Nov 2024



Mass-timber data-center administrative buildings: Aiken, South Carolina completed 2025 (DPR, SmartLam); Cheyenne and Montgomery under way; about 41% less embodied carbon in the materials substituted.

Meta Sustainability, 31 Jul 2025

MASS TIMBER TODAY (CLT, GLULAM)



WITH SUPERWOOD



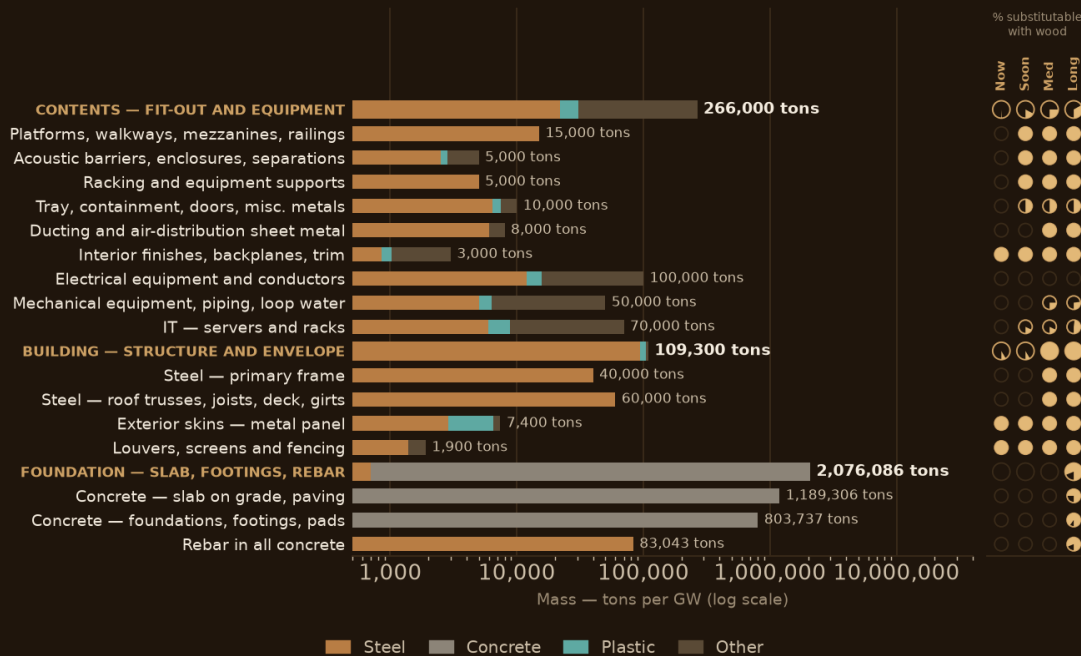
What it replaces	Concrete floor slabs and decks; some columns and beams	Adds the steel: members, skins, screens and fences; later, enclosures and foundations
Strength	Lumber-grade; large sections carry the load	Tensile strength above ASTM A36 steel in samples, at one-sixth the weight; thin, dense members
Enhancing mass timber	Glulam and CLT at lumber stiffness	Mass-timber enhancement: SUPERWOOD outer laminations (~10% of section) raise a glulam beam's stiffness ~75% and strength ~100%
Exterior use	Needs protection from weather	Exterior grade; resists moisture, pests and rot
Fire	Chars; established assemblies	Chars; far better than ordinary wood — Class A demonstrated in testing
Code path	Established in the building codes	Certifiable under wood standards today; SUPERWOOD-specific standards are the goal

Note 4

MATERIALS

What a 1 GW data center is made of

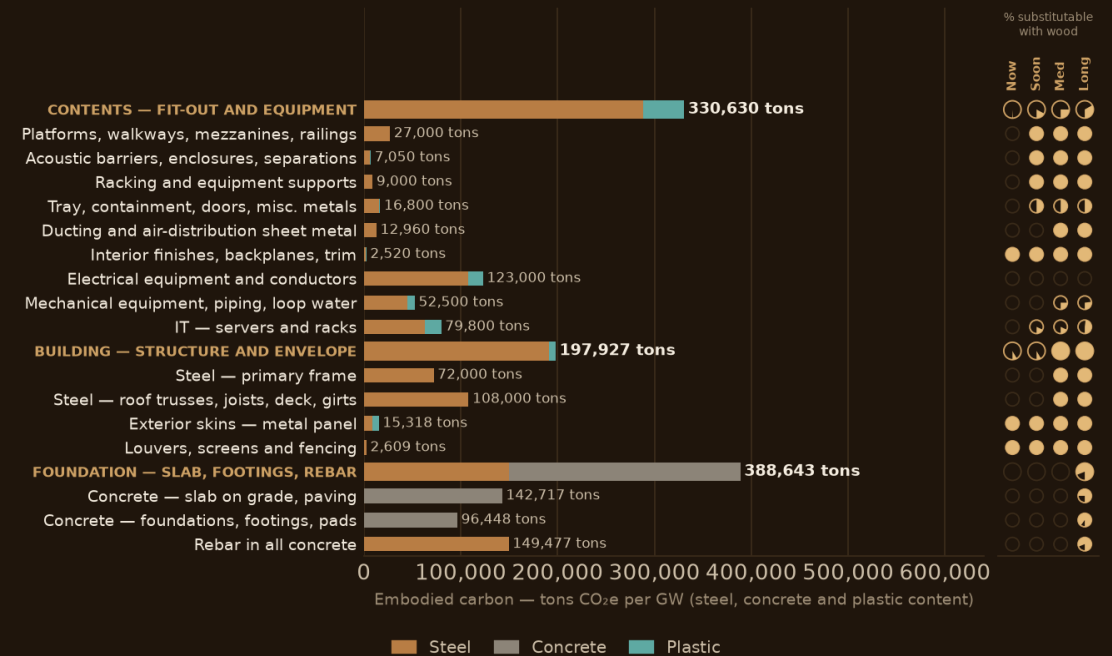
BY MASS



50–80% of the above-ground portion of a data center is steel

Structure, envelope and contents — frame, roof, skins, platforms, racks, enclosures, equipment. Skins, screens and fences now; racks, platforms and barriers next (racks are a few months of development away); frame and roofs as structure; server enclosures eventually; never the electronics.

BY EMBODIED CARBON



By embodied carbon, steel is about two-thirds of everything estimated, including the steel inside the equipment

Building materials alone are about 60% steel at the global-average factor, about a third with recycled steel. The steel and plastics in electrical, mechanical and IT equipment add roughly 250,000 tons CO₂e.

Note 5

Just one data center can consume the *entire output* of a SUPERMILL

SUPERWOOD a 1 GW data center could require, by horizon, against what each mill makes in a year.



Note 6

Properties that matter in a data center



**Stronger than steel.
One-sixth the weight.
Made from wood, in
America.**



STRONG

Able to carry the required loads — tensile strength above ASTM A36 steel in production samples



LIGHT

One-sixth the weight of steel: faster and safer to transport and install, and reduced structural requirements



NATURALLY BEAUTIFUL

The texture, colors and warmth of wood — the biophilic, community-facing surface



DURABLE AND IMPACT RESISTANT

Harder and more dent-resistant than oak; impact and storm resistant; exterior grade resists moisture, pests and rot



INSULATING, NOT CONDUCTING

Thermal and electrical insulator — no thermal bridging, and non-conductive around electrical infrastructure



SOUND AND VIBRATION DAMPING

Particularly for HVAC enclosures, acoustic barriers and equipment screens



RF TRANSPARENT

Useful in niches: antenna screening, timing-antenna enclosures, telecom shelters

Faster to build: lighter, *prefabricated*, reconfigurable

01

One-sixth the weight of steel

Smaller cranes and crews, easier transport, lighter foundations. Time to power is what hyperscalers optimize.

02

Prefabricated to precision

Panels and modules built off site and set like mass timber — standardized designs, plug-and-play systems.

03

Designed for assembly and disassembly

Reusable, reconfigurable components for infrastructure that changes with every hardware generation.

04

Mass timber's schedule record

About 20% average schedule savings across seven case studies, up to 25–30% in others. SUPERWOOD builds the same way.



Concept rendering

Skins, screens and fences from SuperMill One

Every item below ships today as boards up to 8" × 16' × 3/8", exterior or interior grade. SuperMill One makes about one million square feet a year across all markets.

IMMEDIATE shipping from SuperMill One · **SOON** 1–3 years; straightforward applications engineering · **MEDIUM TERM** complex applications engineering such as full building systems, new form factors, or materials engineering · **LONG TERM** 5+ years; technical potential with no design or code pathway yet



Facades, cladding & rain screens



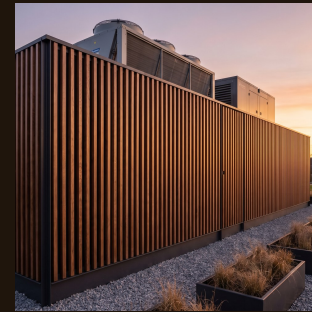
Biophilic interiors for office space



Louvers & equipment screening



Fencing for staff outdoor spaces



Security fencing around transformers & outdoor infrastructure



Trim, door kicks & sub-framing

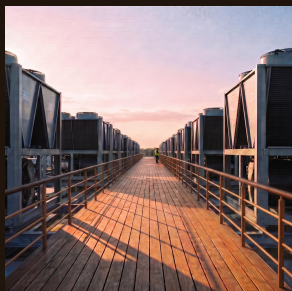
Concept renderings

Why these first. They need no assembly fire rating — where a finish rating applies, Class A has been demonstrated and is specified per order. They sit where the community and the workforce see the data center. And they are the near-term projects our hyperscaler customers asked for: facades, biophilic office interiors, staff-area fencing, security fencing around critical outdoor infrastructure.

Products that need only straightforward engineering



Fan enclosures & ducting



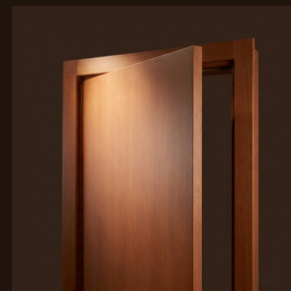
Walkways & platforms



Railings



Window mullions



Interior doors & door
protection



Equipment backplanes



Racks & equipment
supports

Concept renderings

Each needs one scoped test program, which can be co-funded with a customer. Order depends on which a customer picks up first.

Structural: starting *now*, scaling with SuperMill Two

UNDER DEVELOPMENT WITH SUPERMILL ONE OUTPUT

Truss design is starting. Mass-timber enhancement — SUPERWOOD laminations that stiffen and strengthen glulam and CLT-type members — is under way. Light structural components (non-life-safety): strut-type supports, window mullions, deck boards.

We expect structural sales on the way to SuperMill Two. Replacing structural steel is the first thing new data-center customers ask about.

AT SUPERMILL TWO SCALE



Window mullions

Slender, stiff curtain-wall and storefront mullions in place of aluminum and steel — light structural, non-life-safety



CLT-type floor, roof & wall assemblies

Starting with SUPERWOOD hybrid glulam: SUPERWOOD outer laminations on a glulam core. Thin, far stronger panels follow through the mass-timber product route



Trusses, roofs & long-span members

Roof structure first; long-span floors and canopies follow



Heavy equipment supports & server enclosures

Racks come soon, a few months of development; enclosures for servers and equipment follow; the electronics are never replaced

Concept renderings

How qualification works. Every structural application is qualified through a scoped test program with the customer who picks it up first, on the pathway mass timber has already opened in the building codes, with insurer acceptance for structural use. Test methods and property data are available on request.

Prefabricated *envelopes*, then foundations

Prefabricated building envelopes

Hyperscalers are asking for this now, to simplify design and delivery and accelerate schedules.

Structure and skin shipped as panels and modules — the co-development target with our hyperscaler customers' architects and engineers. Shells that carry their own loads, weigh far less than steel and precast, and can be disassembled and moved.

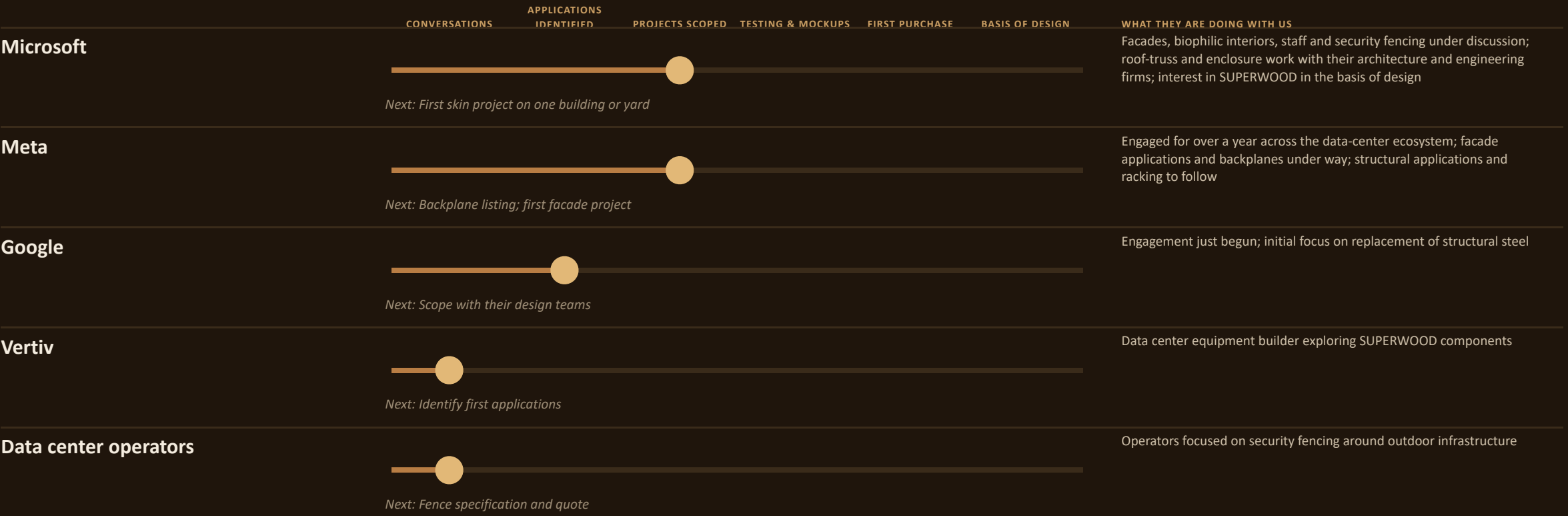
Foundations, slabs and paving

Biggest unaddressed impact both to speed of development and construction, and with huge carbon impact. Geotechnical requirements greatly impact schedule and site selection.

Lightweight, insulated SUPERWOOD foundations and slabs in place of concrete — installed faster, with lower geotechnical demands, and potentially movable.

Stated technical potential, not an engineered plan: slab, paving and foundations are on the order of 1–2 million tons of concrete and rebar per gigawatt. No design or code pathway exists yet.

Customer engagement



ENGINEERING AND CONSTRUCTION COLLABORATORS

Fast+Epp
Top timber engineers

HITT
#1 US data center contractor, ENR 2025

Turner
#1 US contractor, ENR Top 400

Don Davies, advisor
Former president of MKA; structural engineer for several hyperscalers

Note 9

Microsoft and Meta

Microsoft *broad engagement*

Two Northern Virginia datacenters built with cross-laminated timber floor panels on a steel frame — an estimated 35% embodied-carbon reduction vs. conventional steel construction and 65% vs. precast concrete. Design by Gensler; structural engineering by Thornton Tomasetti.

WORKING WITH INVENTWOOD

- **Strategic interest**

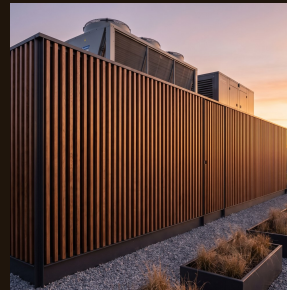
Incorporating SUPERWOOD into the basis of design for future data centers

- **Application areas**

Roof trusses, leading to roofs and fast-to-build enclosures — with their principal architecture and engineering firms

- **Immediate opportunities**

Building facades · biophilic interiors · staff-area fencing · security fencing around transformers



Concept rendering — security fencing around a transformer yard

Meta *engaged for over a year*

Piloting mass timber for data center administrative buildings: first completed in 2025 at Aiken, South Carolina (DPR, SmartLam); projects under way in Cheyenne, Wyoming and Montgomery, Alabama — about a 41% reduction in the embodied carbon of the materials substituted.

WORKING WITH INVENTWOOD

- **Broad engagement**

Across every part of the data-center ecosystem, for more than a year

- **Under way**

Facade applications and equipment backplanes

- **Next**

Structural applications and racking over time



Concept rendering — equipment backplane in an electrical room

Embodied carbon: LCA based on SUPERMILL TWO projections

KG CO₂E, CRADLE TO GATE · EMISSIONS RIGHT OF ZERO, STORAGE LEFT

Steel — global average (BF-BOF), per kg of steel



Steel — recycled (EAF), per kg of steel



SUPERWOOD — per kg of steel substituted at 3:1 to 4:1 replacement by weight, projected: emissions right, carbon storage left



SUPERWOOD carbon estimates are based on the LCA developed by Professor Ming Hu, University of Notre Dame.

Biogenic carbon, reported separately. SUPERWOOD stores 0.5–1.5 kg CO₂e per kg of SUPERWOOD as biogenic carbon, 0.1–0.5 kg per kg of steel it substitutes. Under EN 15804 it is released again in the end-of-life module, so it nets to zero over the life cycle; the lasting advantage is lower manufacturing emissions.

Note 10

Notes and sources

1 Slide 2 • We help data centers decarbonize and improve their impact on communities, while they accelerate SUPERWOOD's journey from premium skins into structural applications

The company, team, mills, cost roadmap and the raise are in the SUPERMILLS Investor Overview. This deck covers one application. Strength: company test data vs ASTM A36; data center sizing: company estimate (slides 5–6).

2 Slide 3 • Industry Growth

US Census Bureau, Construction Spending (C30), private data center construction put in place: July 2026 preliminary seasonally adjusted annual rate \$75.2B vs \$47.8B in July 2025 (+57%); bars are calendar-year totals, not seasonally adjusted; 2026 estimated from the January to July pace; buildings only, servers excluded. JLL North America Data Center Report Midyear 2026 (66 GW). McKinsey, AI power (2025). JLL Global Data Center Outlook 2026. Dell'Oro Group (2026). Steel intensity: arXiv 2509.21312, secondary; tonnage derived. Shortage statements are what customers report to InventWood.

3 Slide 4 • Industry Challenged by Communities

Data Center Watch, Q1 2026 report (June 2026). Virginia HB 153; Henry County, VA ordinance, Aug 2026. Electricity rates and water are the other leading objections; SUPERWOOD does not address them.

4 Slide 5 • Hyperscalers are already building structures with wood. We turbocharge wood.

Sources: news.microsoft.com, Nov 2024; Thornton Tomasetti project page; sustainability.atmeta.com, 31 Jul 2025. Logos and photographs are the companies' own, from the cited publications, used to identify the published projects. SUPERWOOD strength: company test data, parallel-to-grain tension. Hybrid-beam gains: derived from beam theory with SUPERWOOD modulus and strength, engineering write-up pending. Beams: concept renderings.

5 Slide 6 • What a 1 GW data center is made of

Company estimate for a 1 GW IT-load data center, high case. Only the concrete and structural-steel intensities are published (arXiv 2509.21312, a secondary source); other rows are estimates from unit masses. Carbon factors: steel 1.8 kg CO₂e/kg (global average; recycled 0.4–0.7), concrete 0.12; carbon is valued on each row's steel, concrete and plastic content (material split per component: estimates, analyses/material_split.json; polymers at an average 3.0 kg CO₂e/kg); copper, aluminum, electronics, water, gypsum and wood not valued. Concrete: footprint bottom-up with moderate soils (slab, footings, pads, paving, yard mats; the published 500–1,000 m³/MW intensity is the upper comparison). Steel share: company estimate; the 50–80% band covers metal-panel and tilt-up concrete wall designs. Horizons: immediate = shipping; soon = 1–3 years, straightforward applications engineering; medium term = complex applications engineering, new form factors or materials engineering; long term = 5+ years, technical potential with no design or code pathway yet.

6 Slide 7 • Just one data center can consume the entire output of a SUPERMILL

Company estimate from the slide 6 model, low–high scenarios, moderate soils; 0.3–0.6 substitution factor for steel, to be engineering-stamped per element; long-term row also includes server enclosures at 40% of IT mass (a small share of the total), electronics excluded. Mill output at 0.87 kg/sf, 1.3 t/m³. Pricing is set per application; see the SUPERMILLS Investor Overview.

7 Slide 8 • Properties that matter in a data center

Company test data; property data package and test methods available on request. Fire: chars rather than burns, far better than ordinary wood; ASTM E84 Class A demonstrated in testing, specified per order.

8 Slide 9 • Faster to build: lighter, prefabricated, reconfigurable

Schedule figures are mass-timber case studies (ULI Urban Land: ~20% average across seven case studies, 12.7 vs 15.4 months; other studies 25–30%), used as an analogy for a material that builds the same way. No SUPERWOOD schedule data yet.

9 Slide 14 • Customer engagement

Next steps undated. Published backup for Microsoft and Meta on the next slide.

10 Slide 16 • Embodied carbon: LCA based on SUPERMILL TWO projections

SUPERWOOD 0.5 kg CO₂e/kg manufactured and 0.5–1.5 kg/kg biogenic storage: company projections, pre-LCA. Steel 1.8 kg/kg: global BF-BOF average. EAF 0.4–0.7 kg/kg: typical published range for recycled-scrap steel. Replacement 3:1 to 4:1 by weight (0.25–0.33 kg SUPERWOOD per kg steel): company assumption. Reductions are arithmetic on these inputs.

Let's build what's *next*.

- Data centers are short of steel and already build with wood.
- SUPERWOOD turbocharges that wood: skins and fences now, structure in the medium term.
- One basis-of-design win with a hyperscaler is the demand case for SuperMill Two.

Company, mills, cost roadmap and the raise: SUPERMILLS Investor Overview

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